8.1.1 Actions Taken Based on the Results of Evaluation of the COs Attainment (20)

8.1.1 Actions Taken Based on the Results of Evaluation of the COs Attainment

The Department of Computer Science and Engineering follows a systematic Continuous Improvement (CI) mechanism as an integral part of Outcome Course Outcomes (COs), Program Outcomes (POs), and Program Specific Outcomes (PSOs) are periodically analysed to identify gaps and impleme

The department continuously reviews:

- CO attainment analysis
- PO/PSO attainment analysis
- Student feedback
- Employer feedback
- Alumni feedback
- Academic audit reports
- Placement statistics
- Research and innovation indicators

The identified gaps are addressed through:

- Pedagogical improvements
- Remedial classes organised
- Laboratory enhancement
- Faculty development initiatives
- Student support mechanisms
- Industry interaction activities

Process Adopted for CO Attainment Analysis

At the end of every semester, Course Outcome attainment is calculated using direct and indirect assessment tools. The Department Academic Committee (DAC) a target values.

The analysis focuses on:

- COs with lower attainment
- Difficult topics/concepts
- Weak performance trends
- Laboratory skill deficiencies
- Assessment pattern issues
- Teaching-learning gaps

Continuous Improvement Framework

CO Attainment Calculation
↓
Identification of Low Attainment Areas
↓
Root Cause Analysis
↓
Discussion in DAC/BoS Meetings
↓
Corrective Action Planning
↓
Implementation of Improvement Measures
↓
Monitoring and Review
↓
Improved CO Attainment

Sample Analysis of CO Attainment and Corrective Actions

Actions Taken Based on CO Attainment Analysis

Course	Identified Weakness	Root Cause Analysis	Action Taken	Improvement Observed
Computer Architecture	Low attainment in CO3 related to memory mapping and parallel architectures	Difficulty in conceptual knowledge	Conducted additional tutorial sessions	CO attainment improved

Automata	Low attainment in CO2 related to design of automata	Weak in design of turing machine	Conducted additional lecture on turing machine	CO attainment improved
Artificial Intelligence	Weak understanding related to Natural Language Processing	Difficulty in conceptual visualizatio n	Remedial classes on NLP	CO5 attainment improved
Operating Systems	Low attainment in CO5 related I/O, Disc, File system management	Difficulty in understandi ng	Conducted additional tutorial sessions and lab assignments	CO attainment improved
Compiler Design	Low attainment in CO2, CO5 related to symbol table and code optimization and generation	Weak prerequisit e knowledge	Revision classes conducted on automata and formal language	Improved average internal marks
Database Management Systems	Poor performance in CO5 related to design and implementatio n of database	Insufficient hands-on exposure	Added laboratory classes on SQL query	CO attainment improved
Computer Networks	Weak understanding in CO5 and CO6 related to subnetting and network design and implementatio n	Limited practical exposure	Packet tracer based laboratory practice introduced	Improved CO5 performance

Student Support System Improvements

The following support mechanisms were strengthened:

- Faculty mentoring system
- Slow learner support sessions
- Advanced learner enrichment programs
- Coding practice platforms
- Laboratory open access hours
- Technical clubs and student chapters
- Continuous counselling and academic guidance

8.1.2 Actions Taken Based on the Results of Evaluation of the POs/PSOs Attainment (20)

8.1.2. Actions Taken Based on the Results of Evaluation of the POs/ PSOs

Identify the areas of weakness in the program based on the analysis of POs & PSOs attainment levels. Measures identified and implemented assessment years.

POs & PSOs Attainment Levels and Actions for Improvement

POs	Target Level	Attainment Level	Observations
PO1: Engineering Knowledge: Apply knowledge of mathematics, natura science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.			
PO1	2.5	2.63	Target level is achieved. More emphasis must be placed on relating t concepts of science and mathematics to pra applications. The following steps were planned to signific the current Attainment level.
Action 1: Students are strongly encouraged to use mathematical analysis to bet understand and solve engineering problems. Action 2: Assignments were provided to students for a better understanding of engineering fundamentals and an engineering specialization. Action 3: Conducted workshop on "From Probability to Prediction: The Mathema Roots of Machine Learning" on 28th Feb, 2025.			
PO2: Problem Analysis: Identify, formulate, review research literature a analyze complex engineering problems reaching substantiated conclusi with consideration for sustainable development. (WK1 to WK4).			
PO2	**2.5**	2.62	Target level is achieved. The acquisition of problem-solving and anal across first-, second-, and third-year course students to effectively apply them in real-w scenarios. But a higher level of achievemen looked after. Thus, the following actions wer
Action 1: Additional lectures are being conducted to present the core ideas neec to recognize, frame, review, and analyze complex engineering problems. Action 2: To improve their ability to analyze problems, seminars, workshops, ar expert lectures were organized in a variety of engineering fields. Action 3: A research oriented Panel Discussion by the IEEE WIE Chapter was organised on 16th July, 2025. Prof. Dr. Sheli Sinha Chaudhuri, Chair WIE, Kolkata Sectio from Jadavpur University, Prof. Debjani Mitra from IIT(ISM) Dhanbad, Prof. Nandini Mukh from Jadavpur University, Dr. Ranjini Menon, Scientific officer 'F' from VECC, Kolkata, Dr. Manjira Sinha, Senior Scientist from TCS Research & Innovation, Kolkata were present the guest speakers.			

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO3	**2.5**	2.62	Target level is achieved. Different courses, such as DSA, DAA, AI, ar should be followed to design solutions to co engineering problems that involve various a

Action 1: Inspiring students to focus on originality in design while taking into account the relevant safety and environmental standards at both the national ar international levels.
Action 2: Industry-supported training programs were provided in accordance wi memorandums of understanding.

PO4: Conduct Investigations of Complex Problems: Conduct investigatio of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions (WK8).

PO4	**2.5**	2.66	Target level is achieved.

Action 1: One add-on course named "Foundation of Data Science" has been developed, and several projects and case studies are designed to help students become more adept at interpreting and analyzing data.
Action 2: Expert lectures on research methodology will be conducted.
Action 3: Professionally-led seminars on data-gathering strategies, data-preprocessing techniques, and analytical software.
Action 4: Suggested conducting workshops on SPSS, Talend, Tableau, etc.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO5	**2.5**	2.62	Target level is achieved. It has been observed that enhancing tools a resources is imperative to align with industı and facilitate research endeavours.

Action 1: Conducted seminars on cutting-edge technologies in the field of Comp Science & Engineering.
Action 2: Organized seminars on applications of AI/ML in various application are
Action 3: Conducted Webinar / Seminar / Workshop on modern engineering too
Action 4: A Microsoft Azure Fundamentals (AZ-900) Training and Certifica Course was conducted in collaboration with AEM technology from 19.02. to 15.03.2025, and the certification examination was held on 23.04.2025.
Action 5: One-day workshop on Linux Administration and Automation wi on 07.11.2024
Action 6: One-day workshop on Managing Azure Cloud with DevOps Technologies on 12.02.2025

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for impact on sustainability with reference to economy, health, safety, lega framework, culture and environment. (WK1, WK5, and WK7).

PO6	**2.5**	2.74	Target level is achieved. The department proposed courses respond world demands for engineering education tl public health, workplace safety, and ethical considerations.

Action 1: Encouraging students to take part in a variety of extracurricular activit conducted by the department.
Action 2: Conducted a presentation about the state of ethics in the modern wor
Action 3: Organized workshops on current trends of relevance of interdisciplinar courses in societal needs.
Action 4: Student visits to industries that could benefit from engineering advancements are encouraged so that they may gain a better understanding of safety and social implications of these developments. In addition, students are constantly reminded of their societal duties as computer science engineers throu required humanities classes.
Action 5: The students were assisted in their analysis of the projects preliminary independent investigation and its societal impact through the use of a literature survey and actual case studies. Action 6: Students are encouraged to develop projects and guided to highlight the impact in societal and environmental contex
Action 7: In almost all the courses, faculty members take additional classes to s upon 'content beyond syllabus where they give exposure to global development the concerned subject and their implications in economic, environmental, and so contexts.
Action 8: Presented a Seminar on the Long-Term Sustainability of Software Development.
Action 9: Presented a Seminar on "BREAKING FRAMES: On the UN Sustainable Development Goal 5 - Gender Equality" on 17th December 2025.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & internationa laws (WK9).

PO7	**2.5**	2.75	Target level is achieved. Attainment rates are improving and should, further progress, eventually reach the goal.

Action 1: For a comprehensive understanding of the social dimensions, industria visits, industrial training, and internships are organized to enhance students practical knowledge and promote the adoption of improved engineering practice These experiences also contribute to the development of professional ethics, responsibilities, and adherence to the norms of engineering practice.
Action 2: Various programs, such as the National Social Service Scheme, help students develop an understanding of their ethical principles and societal obligations.
Action 3: Expert-led seminars on business ethics for the software professionals.
Action 4: Seminar on Ethical Hacking and Morals for Software Professionals

PO8: Individual and Collaborative Team work: Function effectively as a individual, and as a member or leader in diverse/multi-disciplinary tear

PO8	**2.5**	2.79	Target level is achieved. Attainment rates are improving and should, further progress, eventually reach the goal.

Action 1: In the final years projects, there is a special emphasis on teamwork a collaboration.
Action 2: Students were involved in organizing departmental events.
Action 3: A yearly reunion of the alumni is held, and students are actively encouraged to participate by taking on leadership roles in the planning process. gives them a chance to exercise abilities like leadership and teamwork in both individual and collaborative environments.
Action 4: Students participated on "BLUEPRINT 26, an Inter-College Technical P Presentation Competition" collaboratively on 5th May, 2025.

PO9: Communication: Communicate effectively and inclusively within th engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, mal effective presentations considering cultural, language, and learning differences.

PO9	**2.5**	2.76	Target is achieved. The students abilities in communicating to others, giving presentations, and writing reports will be developed further Attainment rates are rising slowly but steadily, and it is hop future efforts will be sufficient to reach the higher level.

Action 1: Final year students present their software model and solution as a gro enhancing their communication skills and ability to communicate their findings effectively.
Action 2: Conduct soft skill and personality development program classes in the regular session.
Action 3: Emphasis is given on innovative report writing, PowerPoint presentatio with audio video

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO10	**2.5**	2.65	Target achieved. Progressively, attainment rates have been steadily increasing, and it is hoped that continued work wi sufficient to reach the target rate.

Action 1: Presentations are regularly given by software industry experts on the of effective team building and project management.
Action 2: Under the supervision of the project guide, students are supposed to proposals to project financing agencies.
Action 3: Students create a timeline for their courses after conferring with their project guide at the beginning. Project guides keep constant monitoring.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking the broadest context of technological change. (WK8)

PO11	**2.5**	2.64	Target achieved. Projects that are part of the curr have helped the students to perform better in pla interviews.

Action 1: Students are gradually introduced to abilities with long-term benefits through advanced-level training that is likely to remain relevant throughout the careers.
Action 2: Turning their project work into publications has aided them in advanci their higher studies.
Action 3: Students will be encouraged to register for GATE/ GRE/TOEFL and oth competitive examinations.
Action 4: Conducted webinar on "ENIGMA 2025: Explore IoT Through Ha On Learning and Expert Talks" on 16th May, 2025. Prof. (Dr.) Abhijit Debr from UEM University was a speaker there.
Action 5: Conducted webinar on "ENIGMA 2025: Explore IoT Through Ha On Learning and Expert Talks" on 16th May, 2025. Associate Prof. Dr. Su Saha from NIT Durgapur was there as a guest speaker.
Action 6: Students attended the "Finishing School Program on 3D Printi Additive Manufacturing Technology by C-DAC," scheduled for 7th May, 2
Action 7: A career counselling session was held on 25th February 2025, 04:00 PM onwards, at Albert Einstein Hall for pre-final-year students of o college. Mr. Shishir Persai, senior faculty member of MADE EASY, was th key speaker of the session.

PSO 1: Apply principles of computer science to develop efficient compu solutions for complex real-world problems.			
PSO 1	**2.5**	2.62	Target level is achieved. Students can apply what they have learned in Com Science and other basic engineering topics in the pr demonstrated by their final-year projects, participat Hackathons, and feedback from alumni and profess the field. There is a minor deficiency in students theoretical of algorithms, data structures, and problem-solving techniques in complex real-world scenarios.
Action 1: We put more emphasis on the theoretical foundations of Computer Science & Engineering. Action 2: The trade-offs involved in the design decisions of the required final-ye projects were emphasized more significantly. Action 3: Conducted additional problem-solving sessions, coding practice workshops, algorithm design exercises, and case-study-based assignments. Action 4: Organize a coding competition in the tech fest.			

PSO2: Design and develop reliable software systems using modern programming languages, tools, frameworks, and contemporary comput platforms.			
PSO 2	**2.5**	2.62	Target level is achieved. Limited exposure to moder development practices, frameworks, and collaborati development environments. The rate of progress to desired level of achievement is increasing and is on accelerate in the near future.
Action 1: Seminars/workshops on Blockchain, Cyber Security, Machine learning using Python, etc., are arranged on a regular basis. Action 2: Organized industry-led workshops, coding boot camps, software testi sessions, and cloud computing laboratory exercises. Action 3: Students of 6th and 8th semester attended Microsoft Azure Fundamer (AZ-900) Training and Certification Examination with collaboration of AEM technology from 19th Feb, 2025 to 15th Mar, 2025 and the certification examin was conducted on 23th Apr, 2025.			

PSO 3: Integrate knowledge of modern and emerging computing technologies to build innovative and sustainable solutions for societal a industrial needs.			
PSO 3	**2.5**	2.66	Target level is achieved. Limited awareness of susta socially relevant technology solutions.
Action 1: Organized expert lectures, industry seminars, societal problem-solvir projects, and community-oriented technology development activities. Action 2: Interactive Session on Careers in Cybersecurity, AI, and Project Management Certification "Voice of the Students with PMI" was organized on 28 February 2025. Esteemed Guests: Mr. Pallav Rahotgi from Zedroit, Mr. Jaydeep Chakraborty from TCS, Mr. Anutosh Ghosh from GroArche Learning Solutions, M Pratik Kumar Singh from PMI West Bengal Chapter were present as key speake			

8.2 Academic Audit and actions taken thereof during the period of Assessment (15)